L1

CLASS X - SCIENCE

LIGHT

PRASHANT KIRAD

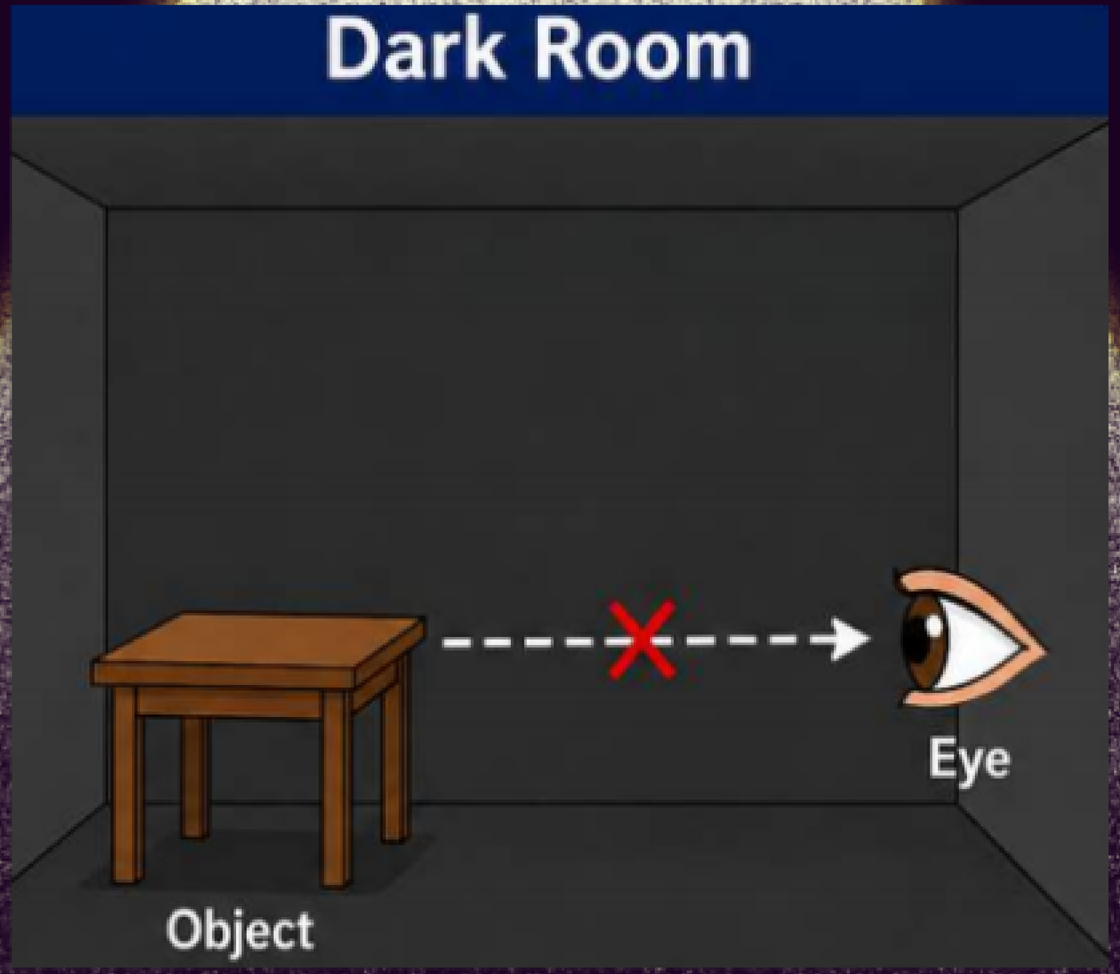
आरंभ BATCH

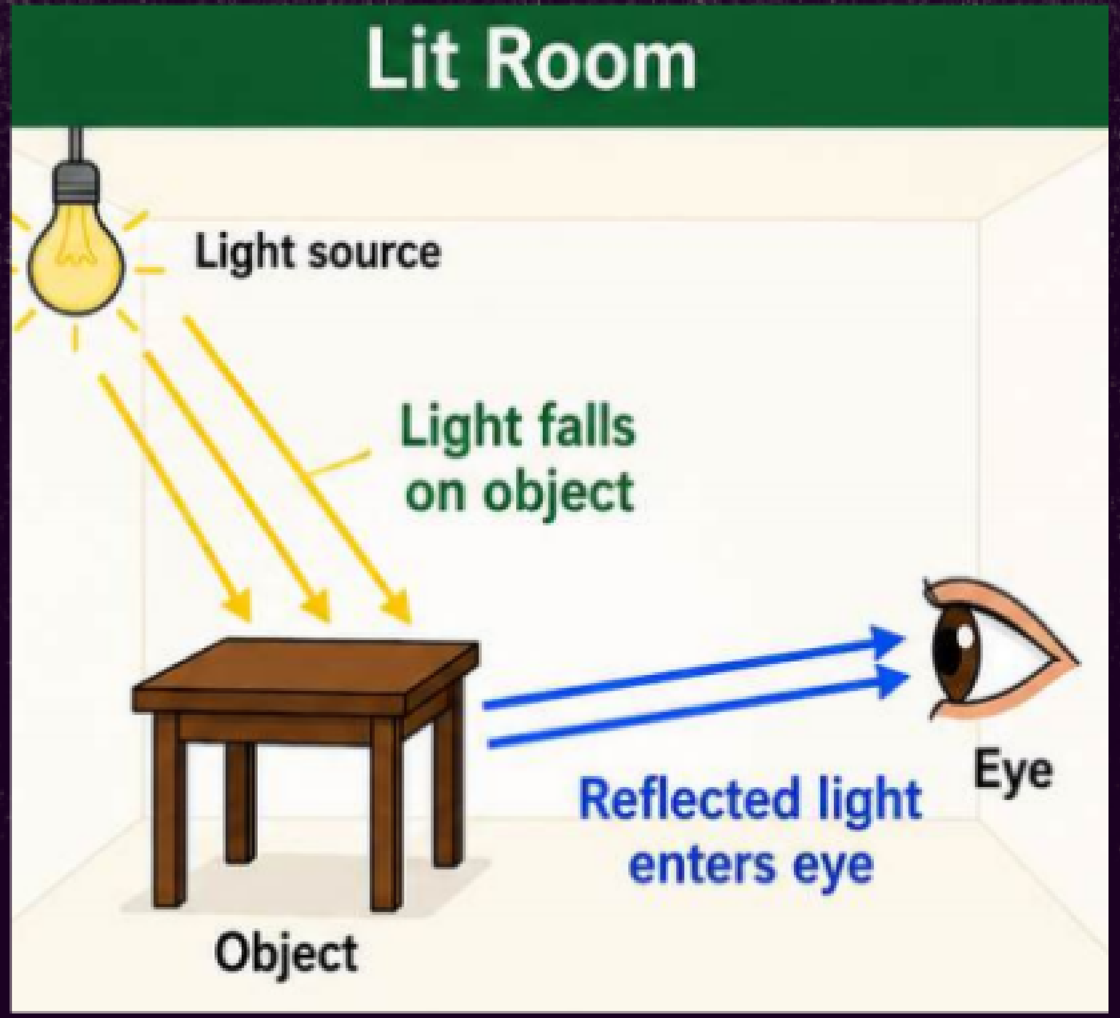
Light is a form of energy that helps us see objects.

Why Do We See Objects?

- We see objects only when light from the object reaches our eyes.
- In a dark room, we cannot see anything because there is no light.
- When there is no light, objects cannot reflect light into our eyes.
- When a room is lit, light falls on different objects.
- These objects reflect the light falling on them.
- This reflected light enters our eyes.
- Because of this, objects become visible to us.

During the day, sunlight is the main source of light.

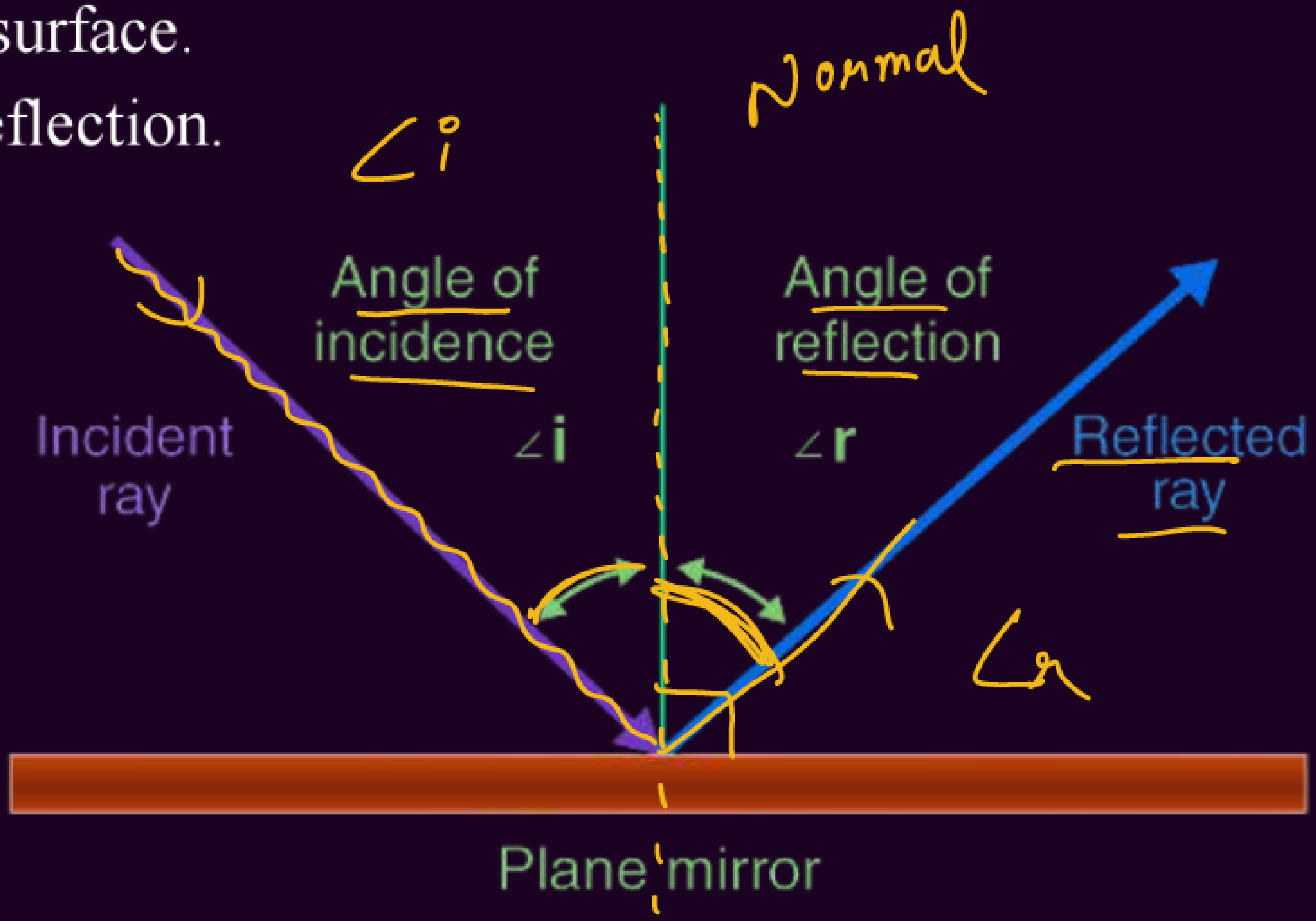




Reflection of Light

Reflection of light is the phenomenon in which light bounces back into the same medium after striking a smooth or polished surface.

- Incident Ray: The ray of light that falls on the reflecting surface.
- Reflected Ray: The ray of light that bounces back after reflection.
- Normal: A perpendicular line drawn at the point where the incident ray strikes the surface.
- Angle of Incidence: The angle between the incident ray and the normal.
- Angle of Reflection: The angle between the reflected ray and the normal.



Laws of Reflection

Q. State the two laws of reflection.

The laws of reflection determine the reflection of incident light rays on reflecting surfaces, like mirrors, smooth metal surface, and clear water.

(i) Angle of incidence (∠i) is equal to the angle of reflection (∠r).

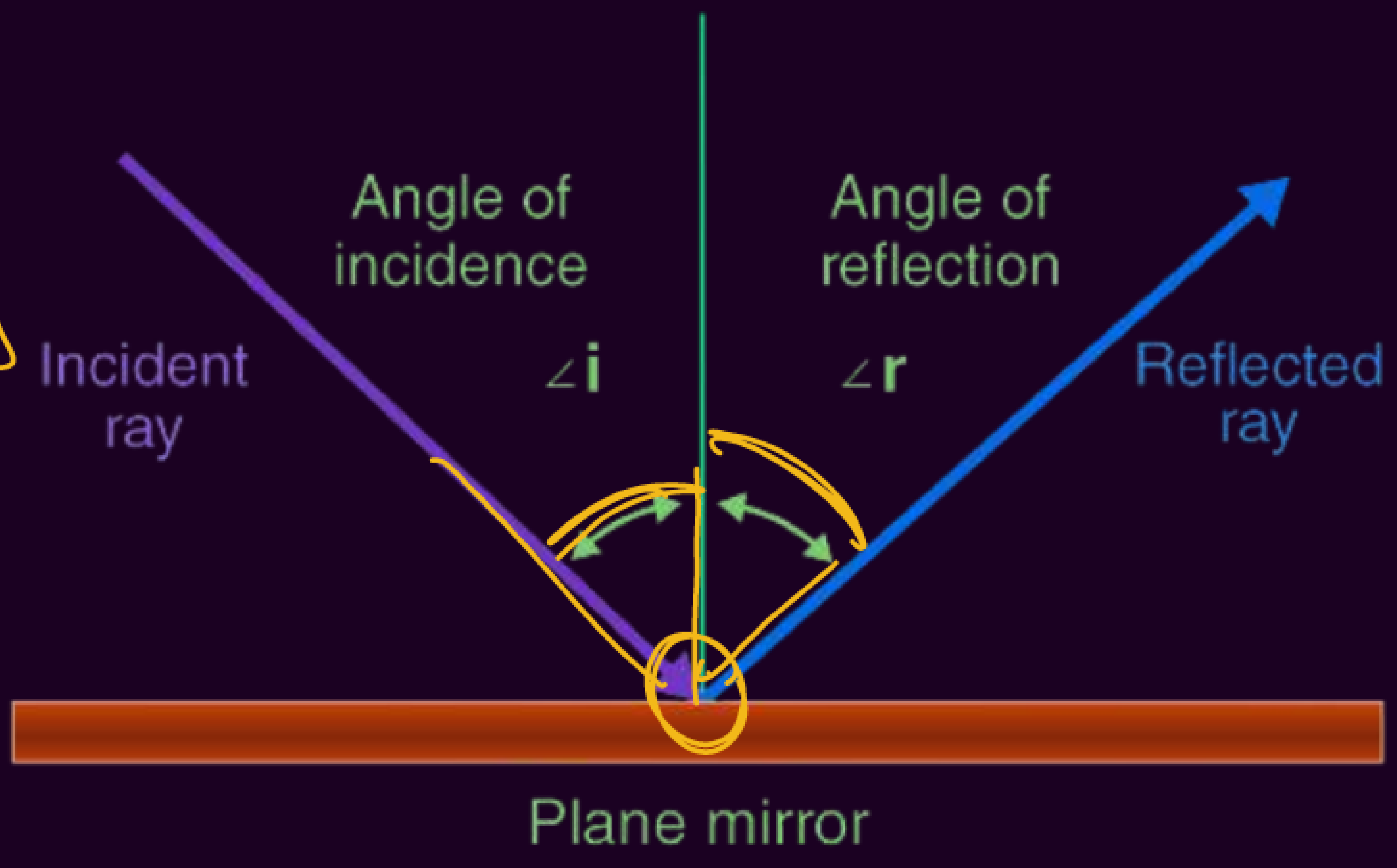
$\boxed{\angle i = \angle r}$

where:

∠i = angle of incidence

∠r = angle of reflection

(ii) The incident ray, the reflected ray, and the normal at the point of incidence, all lie in the same plane.



आरम्भ BATCH

SOCHO!

Q. A student shines a torch on a smooth mirror at an angle of 40° with the normal. What will be the angle of reflection?

A. 30°

B. 40° ✓

C. 50°

D. 80°

$\angle i = 40°$

$\angle r = 40°$

SOCHO!

Q. If a ray of light falls normally on a plane mirror, what will be the angle of reflection?

(a) 0°

(b) 30°

(c) 45°

(d) 90°

SOCHO!

Q. Which statement is true for the reflection of light?

(a) The angle of incidence and reflection are equal.

(b) The reflected light is less bright than the incident light.

(c) The sum of the angle of incidence and reflection is always greater than 90^0 .

(d) The beams of the incident light, after reflection, diverge at unequal angles.

SOCHO!

Q. If a ray of light strikes a mirror at an angle of incidence of 35°, what will be the angle between the incident ray and the reflected ray?

A) 35°

B) 55°

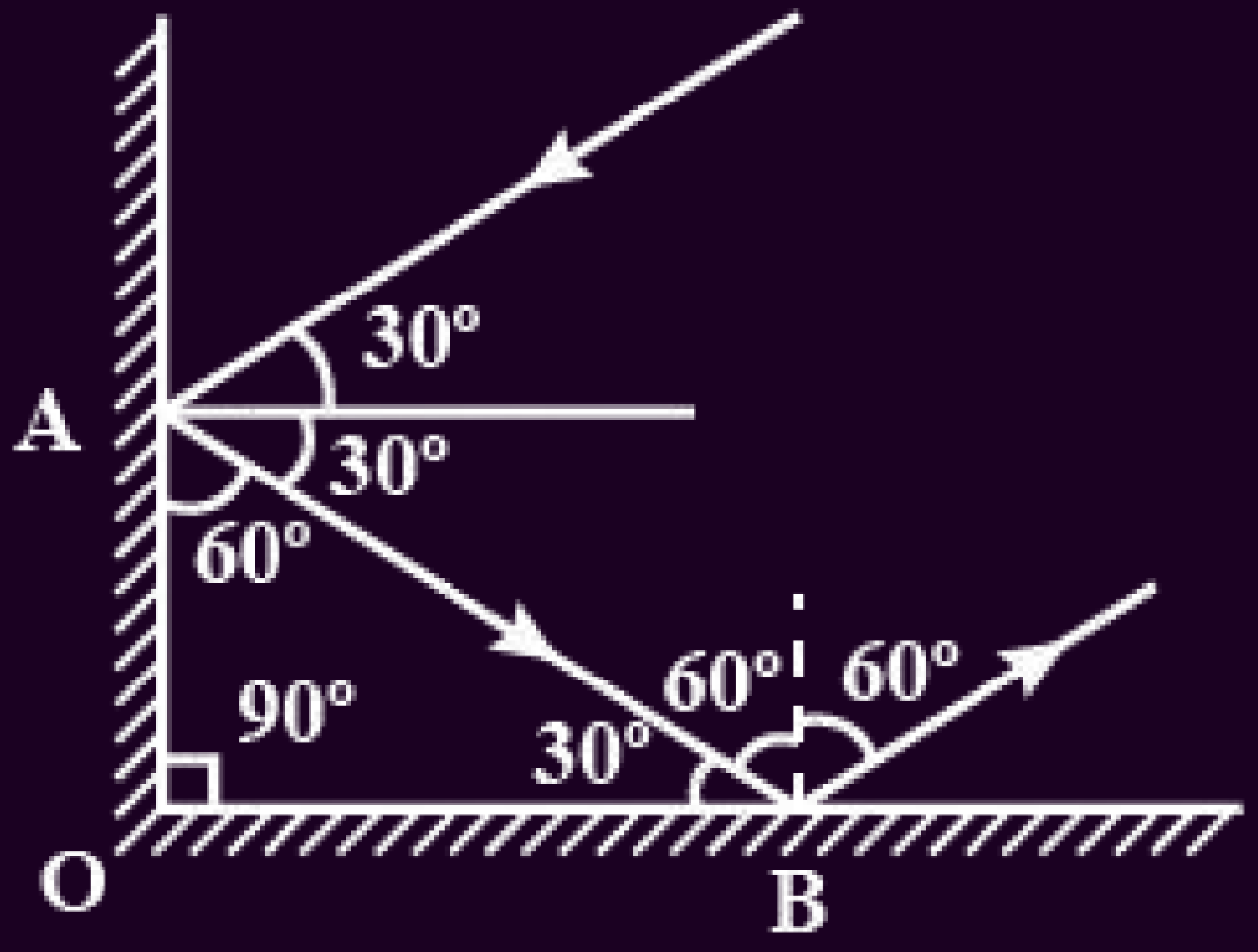
C) 70°

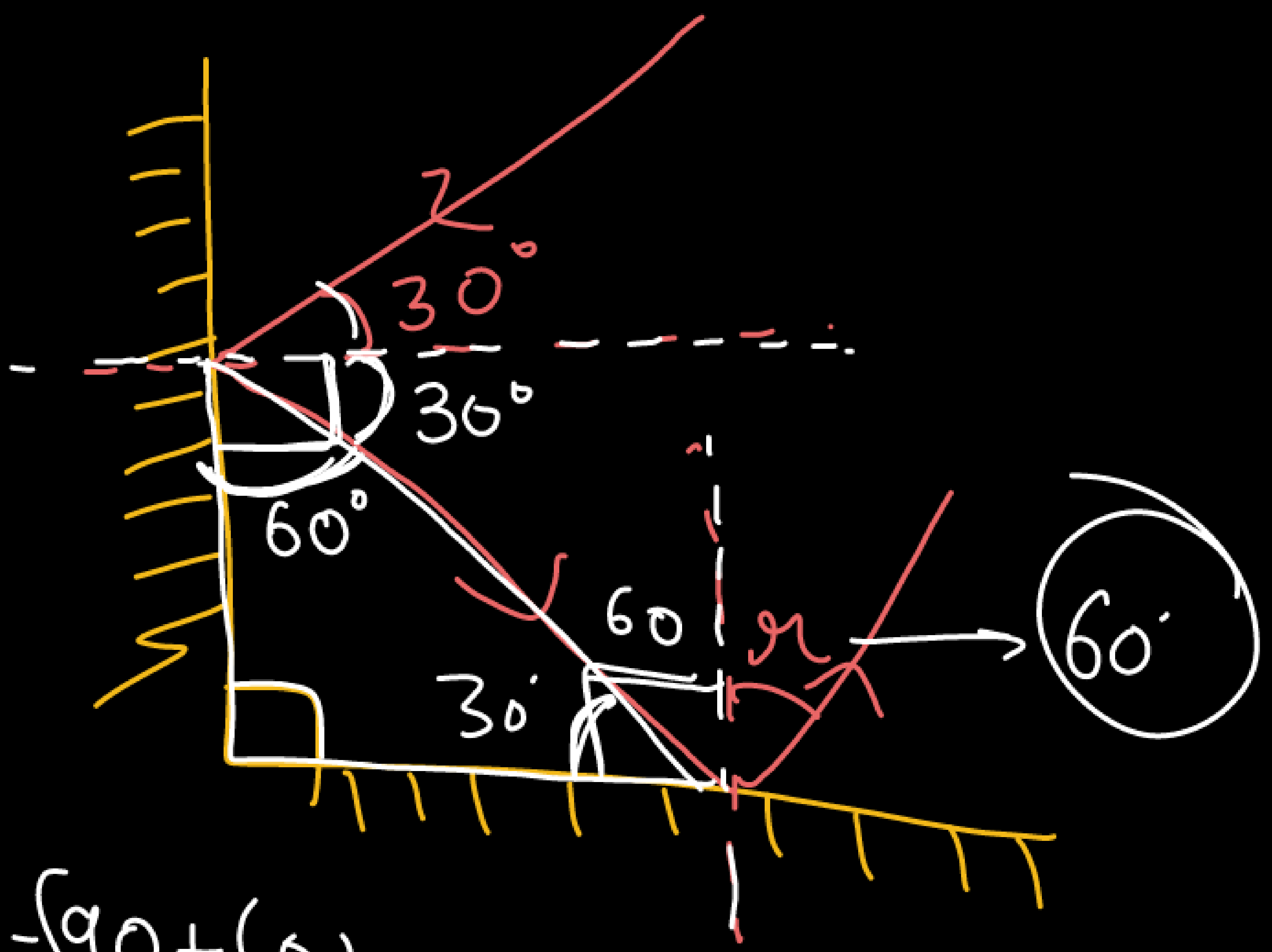
D) 90°

$\angle i = 35^\circ$

Soch ke bataao?

Find the angle of reflection (r) through second mirror.





$$180-(90+60)$$

PLANE MIRROR:

Image Formation by Plane Mirror

Plane mirror: A smooth and polished surface that reflects light uniformly.

- The image obtained is **virtual**.
- The image is **laterally inverted**.
- The image is **erect**.
- The **size of the image is the same as the size of the object.**
- The image formed is as far behind the mirror as the object is in front of it.

Focal length of a plane mirror is infinite

Image

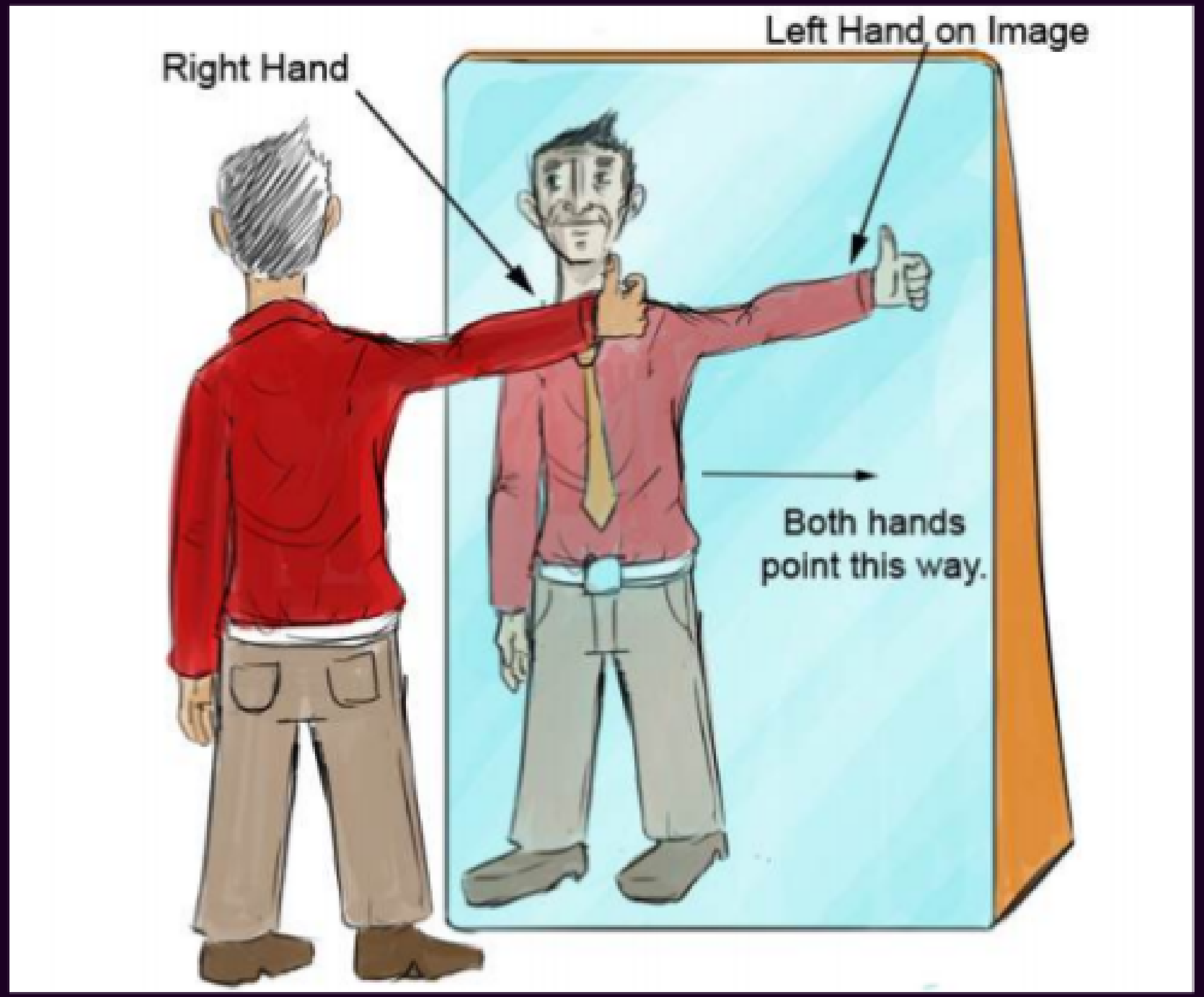
Real → Inverted

Virtual → Erect

↑ → ↑

Lateral inversion Lateral inversion means the left side appears as right and the right side appears as left in the mirror.

Example: If you raise your right hand, the image in the mirror appears to raise its left hand.



Q. Why is "AMBULANCE" written laterally inverted on an ambulance?

That's why ambulances have "AMBULANCE" written in reverse on the front — so drivers see it correctly in their rear-view mirrors!

SOCHO!

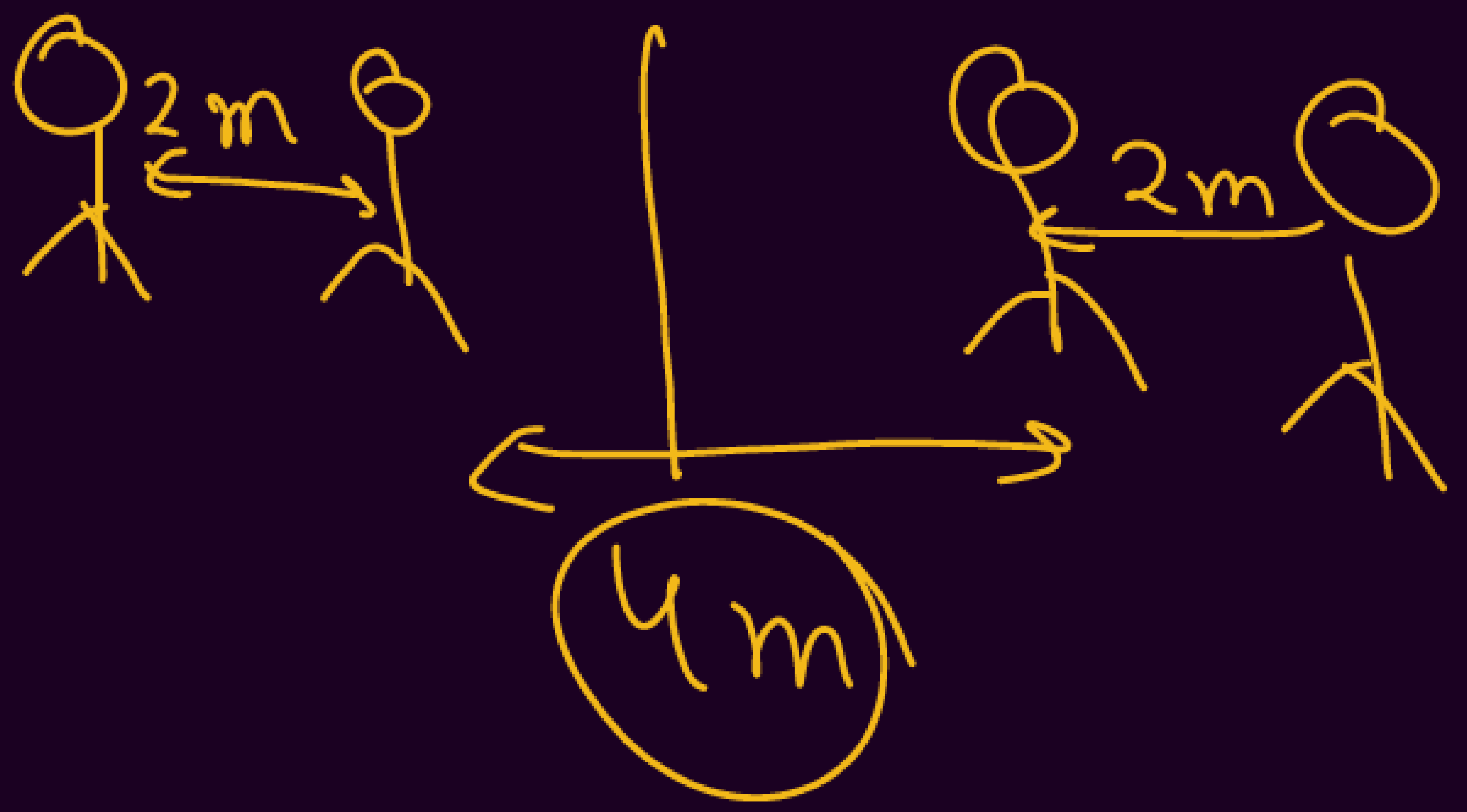
Q. Rahul observes his image in a plane mirror. He moves 2 m closer to the mirror. What happens to the distance between Rahul and his image?

A. Decreases by 1 m

B. Decreases by 2 m

C. Decreases by 4 m

D. Remains same



SOCHO!

Q. Why are mirrors used in periscopes of submarines?

A. To absorb light

B. To refract light

C. To reflect light multiple times

D. To disperse light

NCERT Activity

Activity 9.1

- Take a large shining spoon. Try to view your face in its curved surface.
- Do you get the image? Is it smaller or larger?
- Move the spoon slowly away from your face. Observe the image. How does it change?
- Reverse the spoon and repeat the Activity. How does the image look like now?
- Compare the characteristics of the image on the two surfaces.

Concave Surface (Inner Surface):

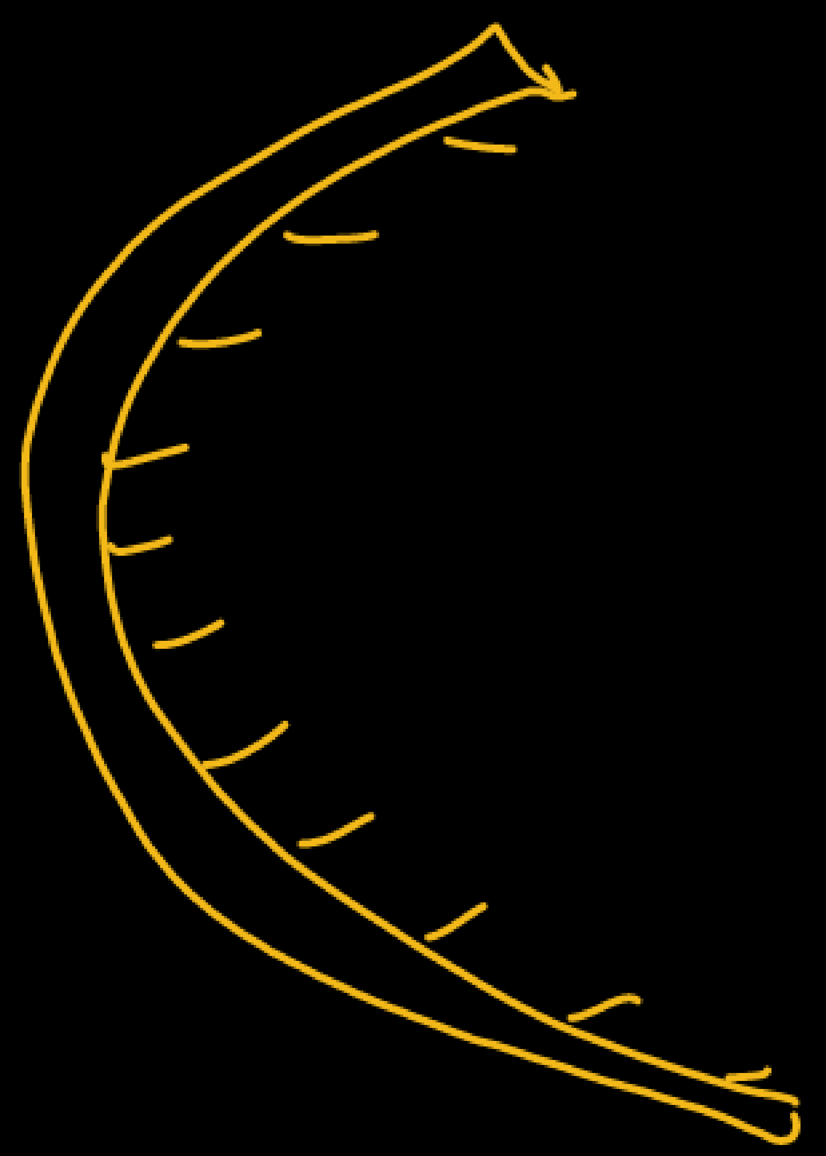
- When the spoon is close to the face, the image appears erect and magnified.
- As the spoon is moved away, the image becomes inverted.

Convex Surface (Outer Surface):

- The image is always erect and diminished.

Concave

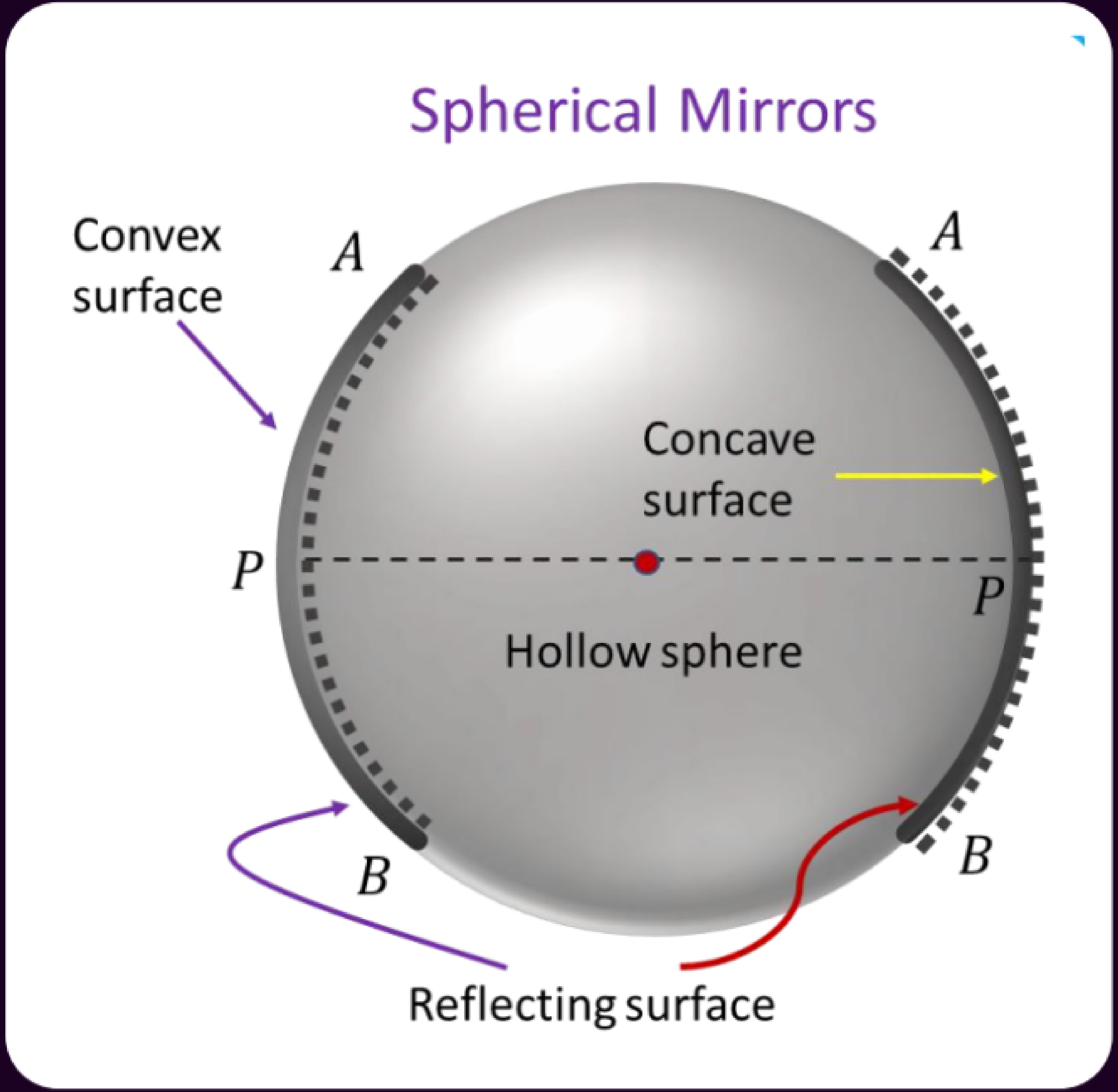
Convex



Spherical Mirrors

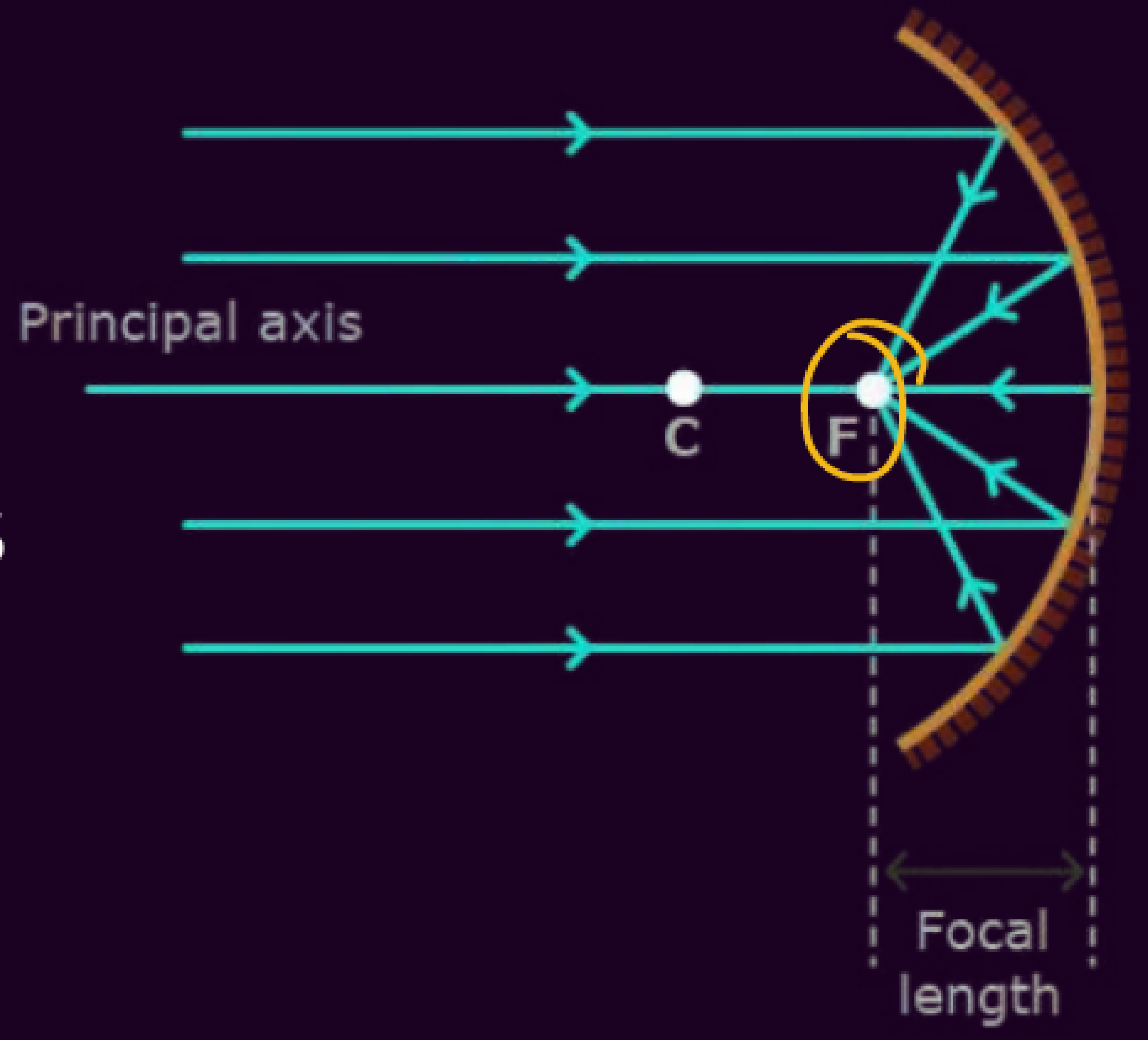
A spherical mirror is a mirror whose reflecting surface forms a part of a sphere.

Its reflecting surface may be curved:
- Inwards
- Outwards



Big + Smaller

Concave Mirror → Converging Mirror



A concave mirror is a spherical mirror that has its reflecting surface curved inward, that is, faces towards the centre of the sphere, like the inside of a bowl.

- It converges light rays that fall on it.
- It is also known as a converging mirror.

Shaving mirrors

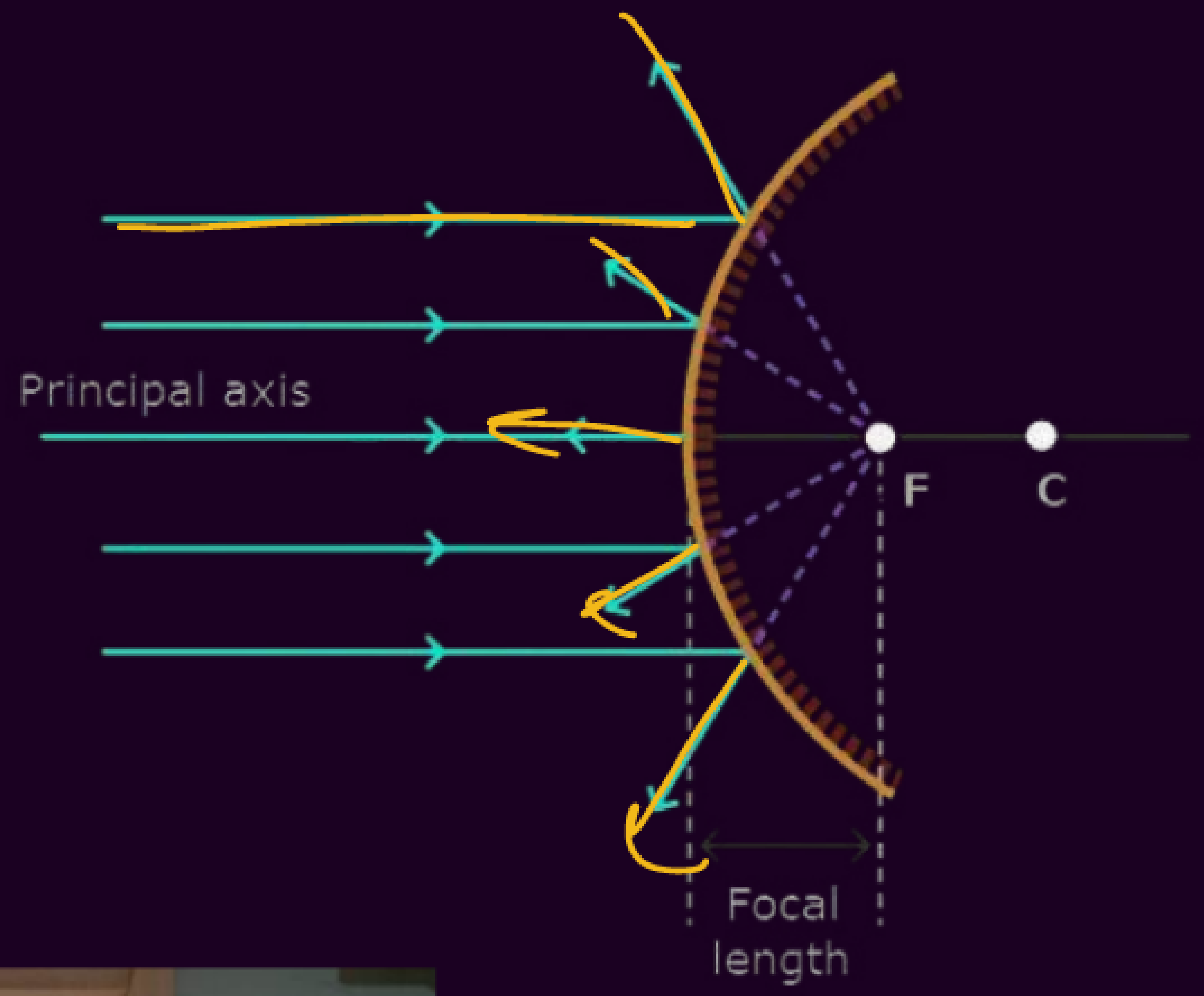
Telescopes

Headlights

Convex Mirror → Small size Image

A convex mirror is a spherical mirror with its reflecting surface curved outward, like the outside of a ball.

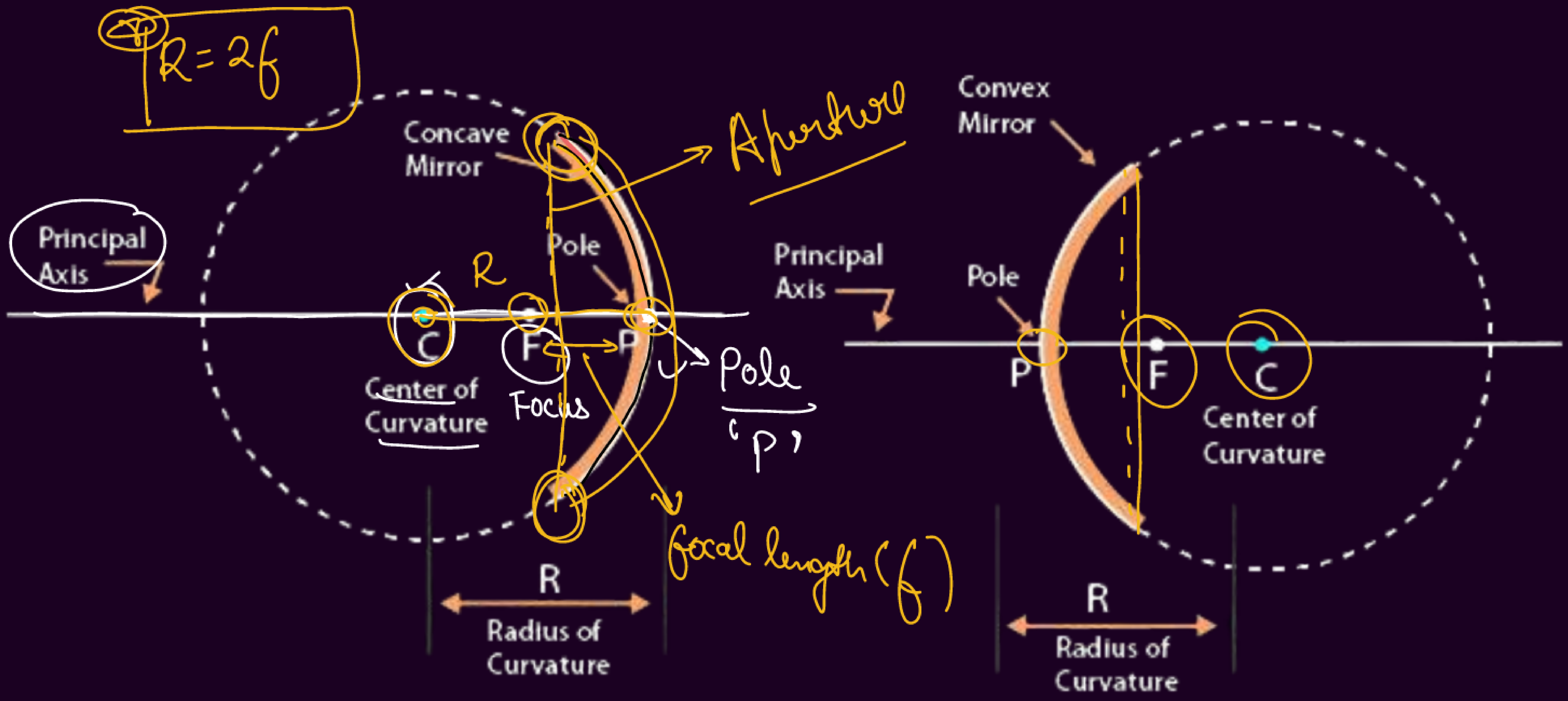
- It diverges light rays that fall on it.
- It is also called a diverging mirror.



Rear view mirror

Security mirror

Spherical Mirrors



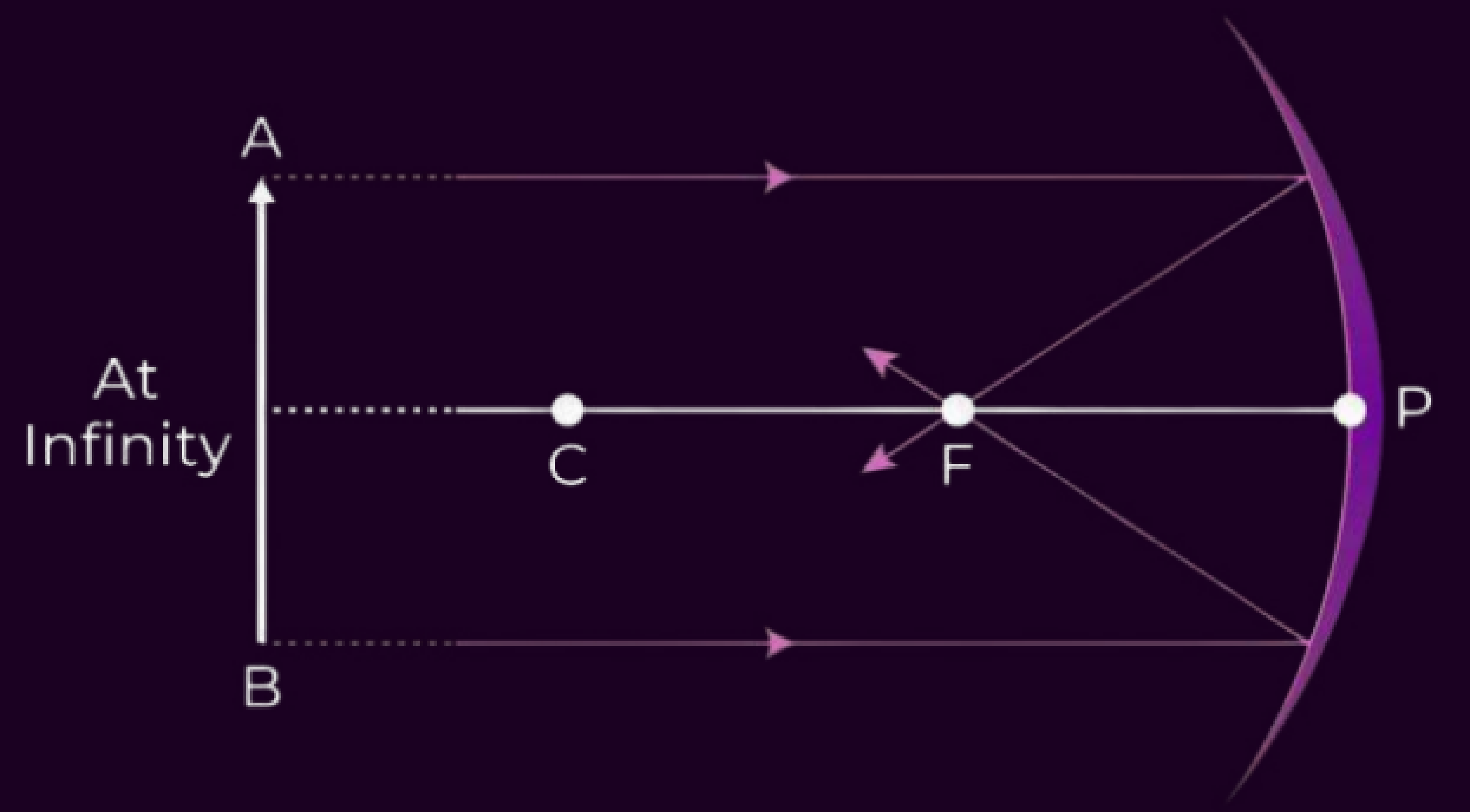
Some Important Terms:

1. **Pole (P)**: The central point of the reflecting surface of a spherical mirror. It lies on the surface of the mirror.

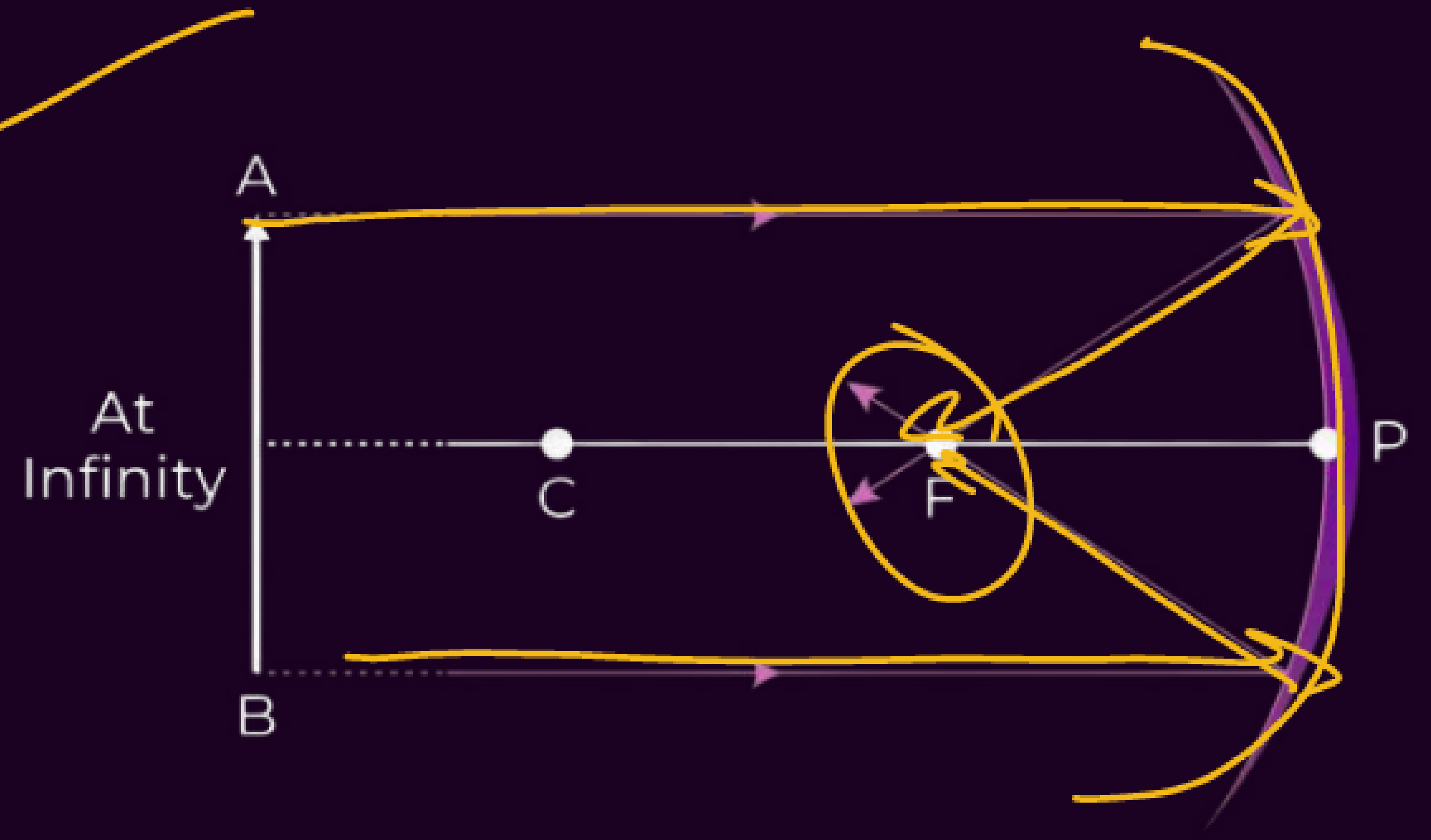
2. **Centre of Curvature (C)**: The center of the imaginary sphere of which the mirror's reflecting surface forms a part. It is not a part of the mirror.

3. **Radius of Curvature (R)**: The radius of the sphere of which the mirror's reflecting surface forms a part.

$R = 2f$



4. Principal Axis: The straight line passing through the pole (P) and the centre of curvature (C) of the mirror. It is normal to the mirror at its pole.



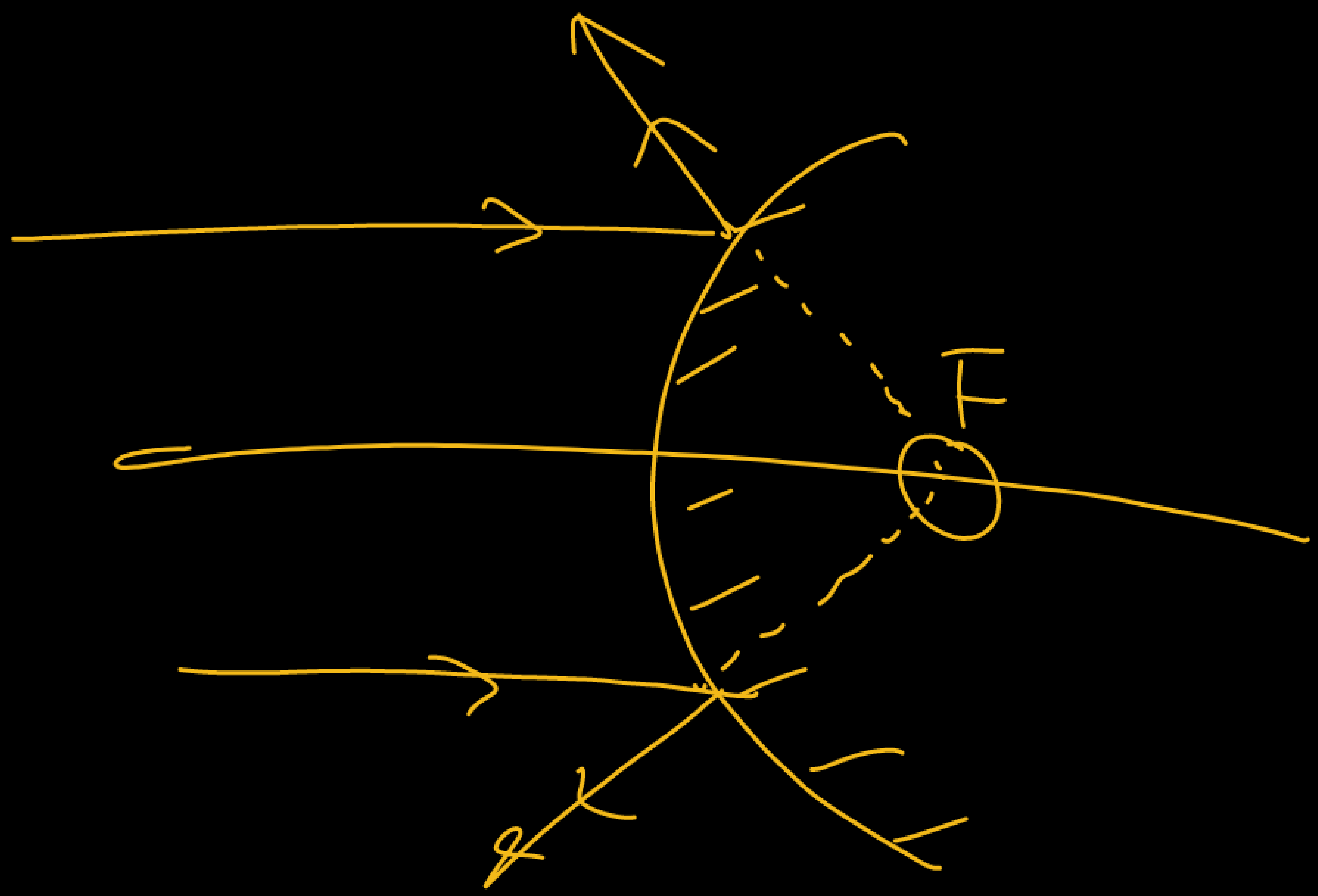
5. Principal Focus (F): The point on the principal axis where parallel rays of light either converge (concave mirror) or appear to ~~diverge~~ (convex mirror) after reflection.

Converge

6. Focal Length (f): The distance between the pole (P) and the principal focus (F) of a spherical mirror.

$$f = R/2$$

7. Aperture: The diameter of the reflecting surface of the spherical mirror


F

NCERT Activity

Activity 9.2

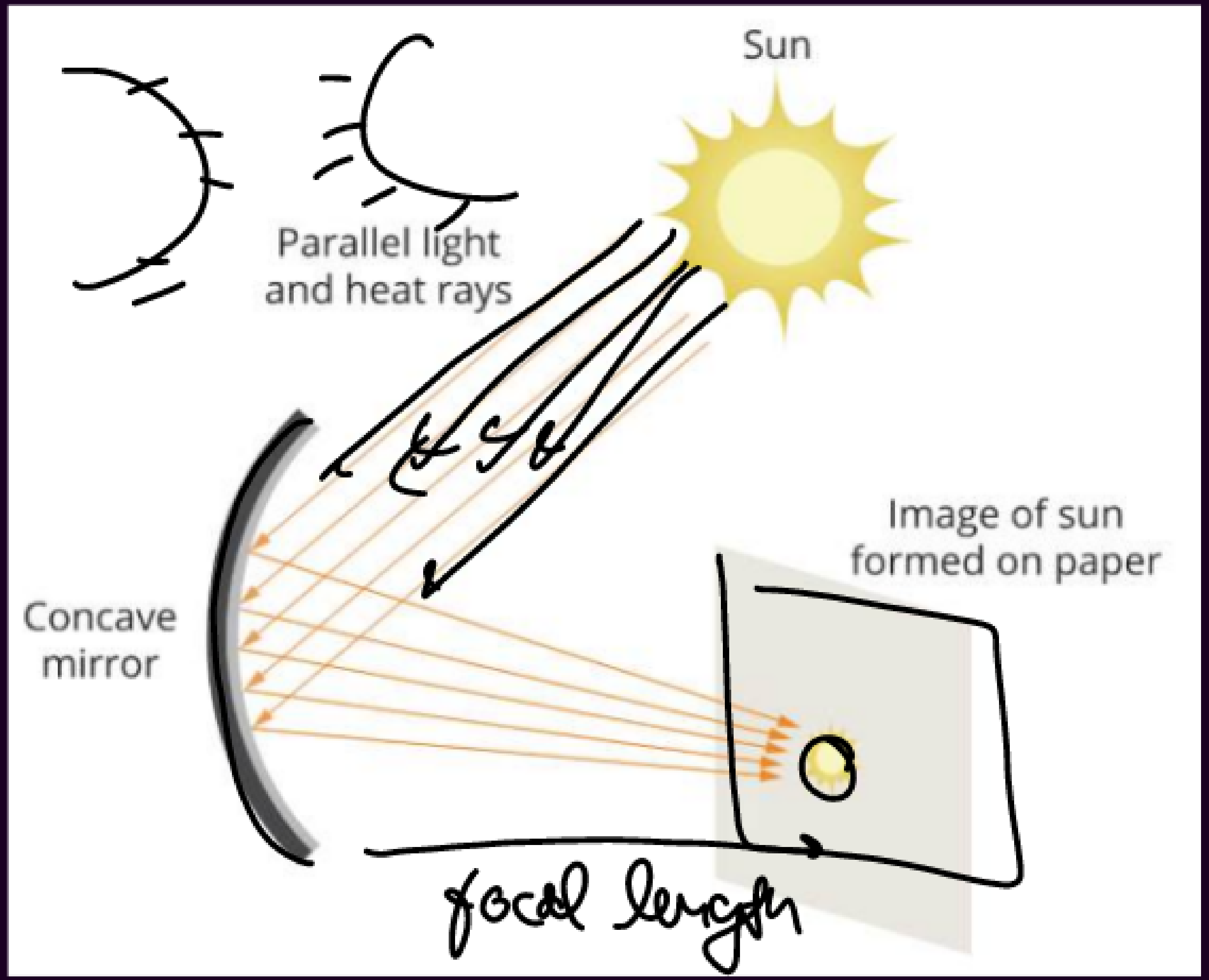
CAUTION: Do not look at the Sun directly or even into a mirror reflecting sunlight. It may damage your eyes.

- Hold a concave mirror in your hand and direct its reflecting surface towards the Sun.
- Direct the light reflected by the mirror on to a sheet of paper held close to the mirror.
- Move the sheet of paper back and forth gradually until you find on the paper sheet a bright, sharp spot of light.
- Hold the mirror and the paper in the same position for a few minutes. What do you observe? Why?

PK Points

Observation:

- A bright and sharp spot of light is formed on the paper.
- The paper begins to burn and produce smoke.
- After some time, the paper may catch fire.



Conclusion:

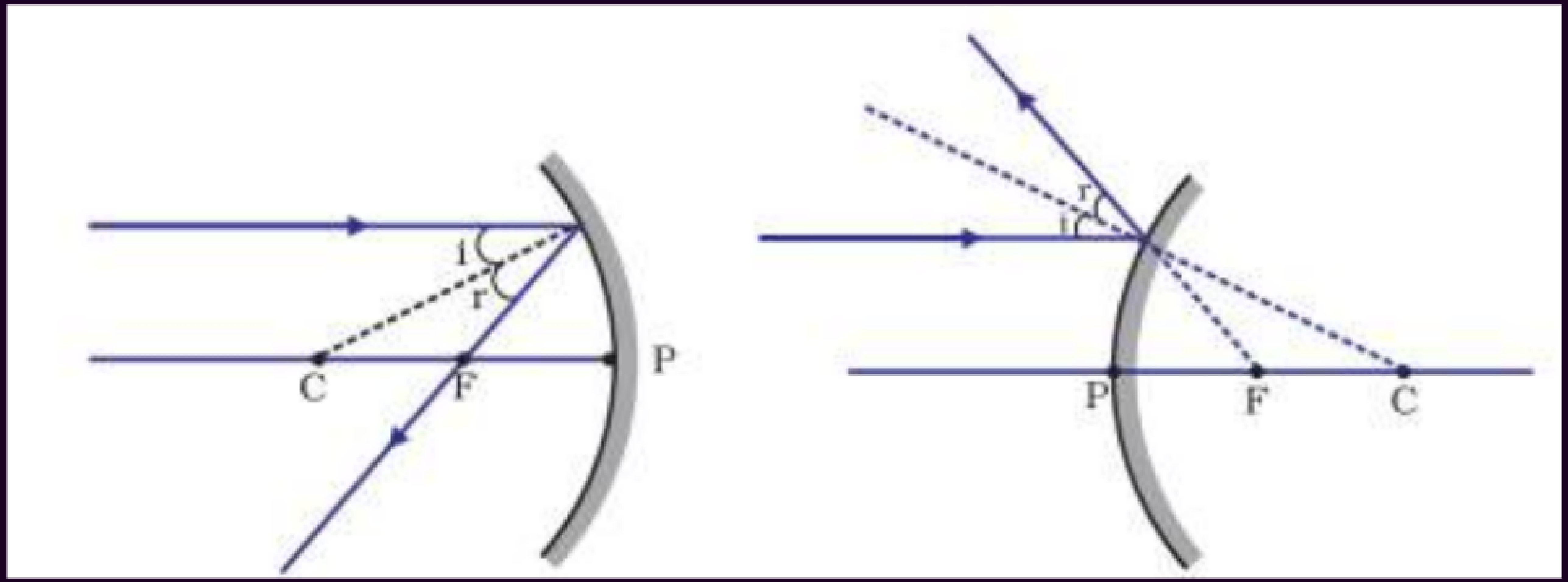
- A concave mirror converges sunlight at one point called the focus.
- The bright spot formed is the image of the Sun.
- Heat gets concentrated at the focus and ignites the paper.
- The distance between the mirror and the bright spot gives the approximate focal length of the mirror.

Rules to Obtain Image:

Rules to Obtain Image:

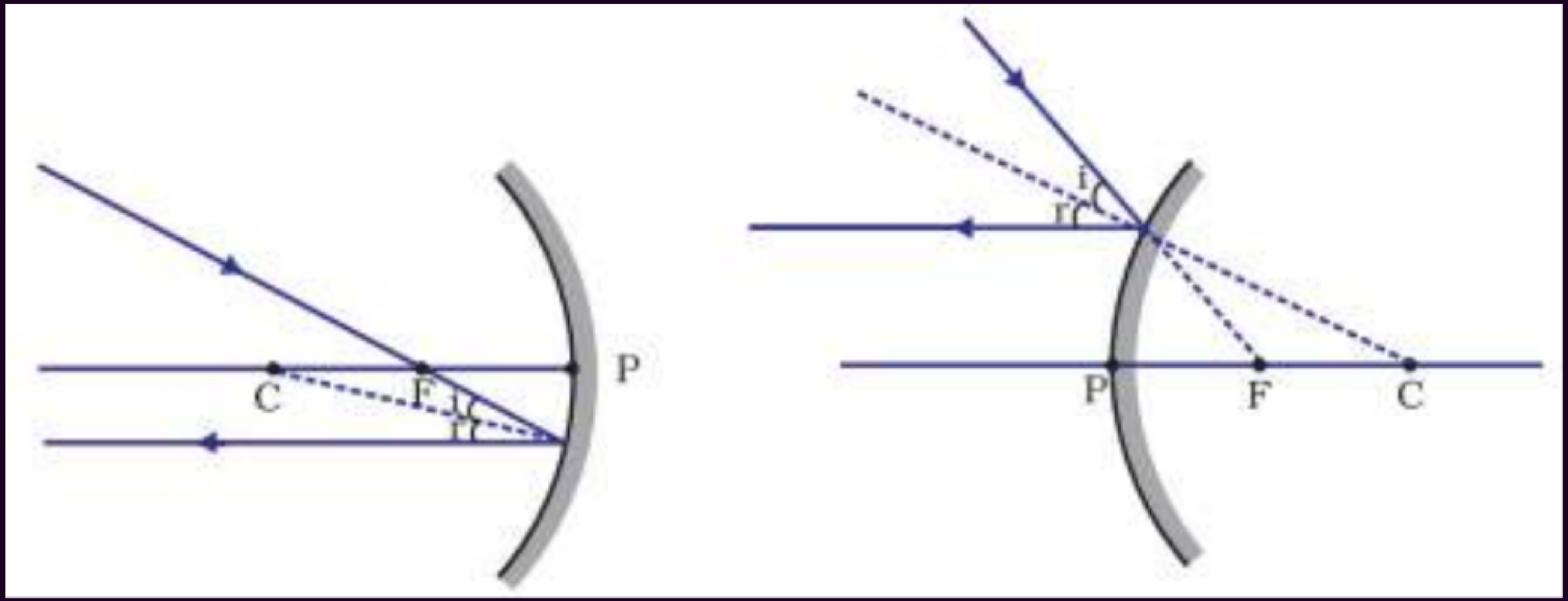
U — Rule

1. A ray parallel to principal axis will pass through focus after reflection.



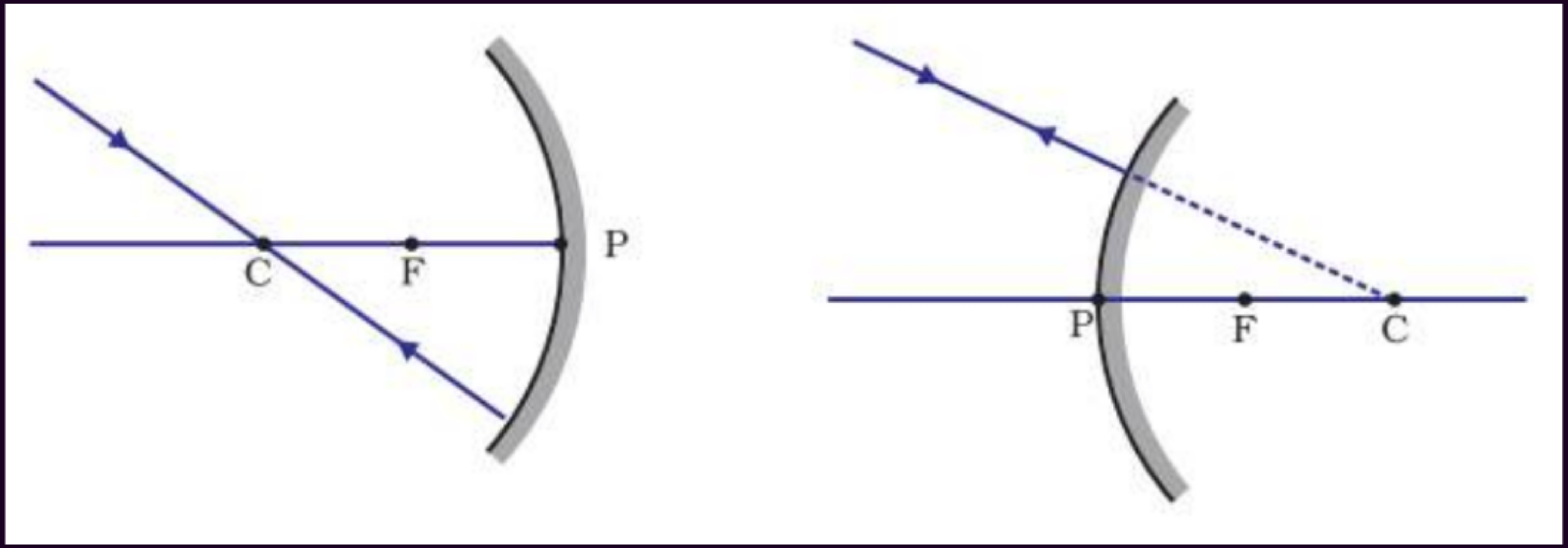
Rules to Obtain Image:

2. A ray passing through the principal focus will become parallel to principal axis after reflection.



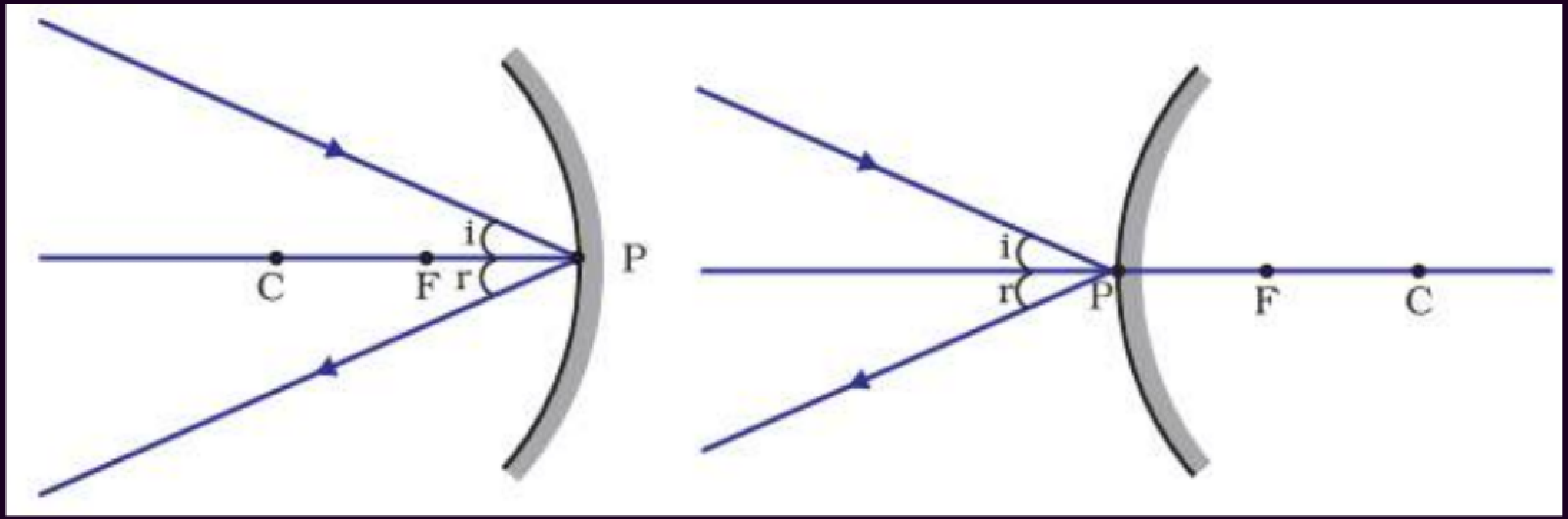
Rules to Obtain Image:

3. A ray passing through center of curvature will follow the same path back after reflection.



Rules to Obtain Image:

4. Ray incident at pole is reflected back making same angle with principal axis.



Kal Milte h!